# Research Statement

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## Overview: Measuring Representation, from Ballots to Language

Societies increasingly make collective decisions through algorithms. Cities allocate budgets with voting rules, parliaments are elected under competing electoral systems, and large language models now summarize public consultations and draft consensus statements. Each of these raises the same question: *does the outcome faithfully represent the people it claims to represent?* Answering it requires a science of *measurement* for collective decisions: principled ways to compare decision problems, to quantify the structure of preferences, and to evaluate outcomes against real data. Until recently, computational social choice had few such tools: it was a largely theoretical field, with scarce real data and experiments run on votes drawn uniformly at random. My work has changed this. I introduced the measurement tools, built the datasets, and helped set the experimental standards that the field now uses. At EPFL, I will take the next step: **measuring and guaranteeing representation as collective decisions move from ballots to language.**

I bring three assets to this program. I built each of them, and each is now used by others:

- *A geometry of collective decisions.* I introduced *maps of elections*, based on distances that measure how similar two decision problems are, independently of who the candidates and voters are. The founding paper has over 60 citations, the line of work has been cited in more than 120 papers, over 50 of them by researchers outside my circle of coauthors, and reviewers now ask authors to use diverse data "in the times of the map of elections."
- *Real data at scale.* I initiated *Pabulib*, the largest open repository of participatory budgeting data (2,053 elections from 30 cities in 8 countries, 8.6 million votes). Nearly all experimental research on participatory budgeting now uses it, and its analyses helped convince cities in Poland and across Europe to adopt the Method of Equal Shares.
- *A channel to practice.* I co-founded *Science for Democracy*, a nonprofit that helps municipalities adopt proportional voting rules.

I have led this agenda from its start. As a PhD student, I was the principal investigator of a National Science Centre grant (NCN Preludium) on the *election isomorphism problem*, which grew into maps of elections. The positionwise distance at the core of the framework was my own idea, I designed and implemented all the experiments and software behind it, and I am lead author of its central papers (AAMAS'20, IJCAI'22, AIJ'25). My thesis, the first in the field to take the whole space of elections as its object of study, was defended with distinction at AGH University in 2024 and received the Polish Prime Minister's Award (2025); my doctoral work also earned a START fellowship of the Foundation for Polish Science (2023). Since then, a CNRS postdoc at LAMSADE (Paris Dauphine-PSL) and my position at the University of Geneva have taken this program into the domain of language.

## Past and Current Research

**How similar are two elections? A geometry of collective decisions.** Two elections may have entirely different candidates (presidential candidates vs. types of sushi), so a meaningful distance between them must be invariant to renaming candidates and reordering voters. In my NCN-funded project, we showed that the most natural such *isomorphic distances*, such as the number of adjacent swaps needed to make two elections isomorphic, are NP-hard to compute, hard to approximate, and hard in the parameterized sense; even CPLEX and Gurobi fail on them (AAAI'19; JCSS'25; AAAI'22; JAIR'24). My answer was the *positionwise distance*: represent

each election by its positional matrix (how often each candidate takes each rank) and compute the optimal matching of candidates under the $\ell_1$ norm. It runs in polynomial time and is highly accurate in practice, and it let me draw the first *map of elections*: compute pairwise distances among a collection of elections and embed them in the plane so that similar elections lie close to each other (AAMAS'20). We then gave the map interpretable reference points (IJCAI'21), extended it to approval elections, in a paper on statistical cultures for approval voting that the community now treats as the reference (IJCAI'22), to elections of different sizes through *feature distances* (AAAI'25), and to preference learning: through the geometry of expected frequency matrices, we learn the parameters of the Mallows model in polynomial time, where earlier Kemeny-based approaches were NP-hard (NeurIPS'22). The full framework appeared in *Artificial Intelligence* (AIJ'25).

Maps turned out to be a methodology, not just a picture, and they corrected how the field ran experiments. They revealed that the Mallows model, the most popular model of random votes, had been used for years without normalizing its dispersion parameter by the number of candidates, so that its instances crowded into a narrow region of the space; the normalization I proposed is now standard. They showed that the running time of ILP solvers for NP-hard rules grows with an election's distance from impartial culture, and that a family of group-separable elections that no earlier paper on Dodgson's rule had considered is especially hard for it. They also showed how little of the space the available real-life elections covered, which motivated others to collect more: Boehmer and Schaar (AAMAS'23) added nearly 8,000 instances to PrefLib so that they could be placed on the map. I extended the framework to stable roommates and stable matching instances, in a paper with two fellow PhD students and no senior coauthor (AAMAS'23 *Best Student Paper Award*; JAIR'24), and later to tournaments, fair division (AAAI'26), and voting rules themselves. To make the methodology standard practice, I co-authored a community guide (IJCAI'24), gave tutorials at AAAI, AAMAS, and IJCAI, and built two open-source Python libraries: `mapel`, for drawing maps, and `prefsampling`, for sampling synthetic preferences.

**Discovering structure in preferences.** Beyond comparing elections, I develop measures of what is *inside* them. Polarization has long been studied in political elections where voters name one party, but was almost entirely neglected for ordinal and approval ballots. We introduced indices of agreement, diversity, and polarization built on a single generalization of Kemeny's rule, *k-Kemeny*, which seeks $k$ representative rankings instead of one consensus; the problem is NP-hard, and we gave efficient approximation algorithms and extended the indices to approval elections (IJCAI'23; IJCAI'26), then used them to quantify how diverse structured preference domains are (AAAI'26; AAMAS'26). During my postdoc with Jérôme Lang, I initiated and led a line of work that inverts the usual direction of the field: instead of designing a rule, we *discover hidden properties of a preference dataset*, such as clones, antagonistic candidates, and diverse groups of candidates (AAMAS'24; IJCAI'24), and, in ongoing work, hidden structure among voters, such as clusters of similar voters and outliers. This brings social choice close to the central task of machine learning: finding structure in data. Closer to political science, I designed voting rules that let allied candidates avoid the spoiler effect (AAMAS'24) and a nonparametric test for gerrymandering (IJCAI'24).

**Real data and real deployments.** In participatory budgeting (PB), residents vote on how to spend part of a city's budget. In 2020 I initiated *Pabulib*, with D. Stolicki and N. Talmon, and I have led its development since (IJCAI'23). It now holds 2,053 elections from 30 cities in 8 countries, with 8.6 million votes on over 38,000 projects; researchers abroad now persuade their own cities to contribute, which is how Amsterdam and Budapest entered it. Pabulib is the standard dataset for PB research and, increasingly, for approval voting in general: more than 70 papers have used it, including field studies of participatory processes in Switzerland and Taiwan. Real data opened questions that theory alone could not answer: learning realistic

models of approval voting (AAMAS'25), measuring how well individual projects performed under a given rule (IJCAI'24), and a game-theoretic analysis of how proposers set project costs under the Method of Equal Shares (NeurIPS'25).

This work reaches practice. The analyses in the Pabulib paper were used by the team, of which I am a member, that convinced cities in Poland and across Europe to adopt the Method of Equal Shares, starting with Wieliczka, the first city to use it. Proportional rules sometimes fund a costly project over a cheaper and more popular one, which is correct but surprising to residents, so I have also worked on ways of presenting outcomes that explain such decisions. In 2025 I co-founded *Nauka dla Demokracji* (Science for Democracy), a nonprofit that connects researchers with the officials who run democratic processes. Pabulib now operates under its auspices, closing the loop between the data that cities produce and the research that improves their processes.

**Toward language.** At CNRS LAMSADE, I built simulation frameworks for multi-agent decision-making. At the University of Geneva, I build NLP pipelines that extract structured information from treaties, diplomatic communications, and policy documents, and I model decision-making dynamics in multilateral negotiations. Bringing these together, in *VoteSim* (EMNLP'26) we simulate democratic deliberation among LLM agents and show how the choice of electoral system shapes whether pluralistic preferences are represented.

## Research Agenda

Collective decisions are moving from ballots to language. Residents submit free-text proposals, consultations collect thousands of comments, and AI systems summarize, mediate, and increasingly vote on people's behalf. Recent work has made a promising start. *Generative social choice* (Fish et al., EC'24) gives proportional-representation guarantees for slates of LLM-generated statements, and the *Habermas Machine* (Tessler et al., *Science* 2024) showed that LLM-mediated deliberation can find common ground at scale. Three things are still missing: ways to *measure and audit* representation in text, an understanding of the *dynamics* that arise when AI agents take part repeatedly, and *validation on real data* rather than synthetic opinions. These are exactly the gaps that my work closed for ballots. My agenda has three connected thrusts; each poses concrete open problems and has a natural first PhD project.

### Thrust A: Representation guarantees for collective decisions in language.

- **Auditing representation in text.** Given individual opinions written in natural language and an aggregate text (a summary or a consensus statement), how should we measure how proportionally the aggregate represents the population? Which measures satisfy natural axioms, and which can be estimated with few LLM queries and provable error bounds? Existing methods *generate* representative text; we lack principled ways to *evaluate* it. My structure-discovery methods, which find clusters and outliers among voters, are the natural starting point for finding the groups that an aggregate text leaves out.
- **Polarization dynamics in AI societies.** Under which electoral systems, discussion structures, and mixes of model families does repeated deliberation among LLM agents converge, fragment, or polarize? I conjecture that majoritarian systems accelerate polarization and that diversity of model families slows it. My polarization indices provide the measures, and records of real multilateral negotiations provide a human baseline.
- **Faithful AI proxies, tested on real ballots.** Pabulib contains real approval ballots over projects with textual descriptions, which is rare ground truth for AI proxies. When an LLM votes on a person's behalf, how much does the *outcome* change under rules such as the Method of Equal Shares, and which rules are robust to proxy errors? This is the first PhD project of the thrust.

*Five-year goal:* a theory of representation for text-based collective decisions, with audit tools that

practitioners can apply to AI-generated summaries and consensus statements.

**Thrust B: Participatory democracy as a living laboratory.**

- **Richer democratic data.** I will extend Pabulib with full ballots, project texts, residents' comments, and outcomes over several years. This will turn it into a longitudinal benchmark, including for Thrust A.
- **Does the rule change the process?** Theory predicts that proportional rules change which projects are proposed and at what cost (NeurIPS'25). Field studies with partner cities can test whether they also change who participates and who proposes. This is the first PhD project of the thrust.
- **Tools that cities and residents can trust.** Open tools to choose, simulate, and audit PB rules, and to *explain* their outcomes to residents, so that cities can adopt evidence-based designs without in-house experts and residents can see why a project was or was not funded.

Switzerland is an ideal setting for this work, with its tradition of direct democracy, Lausanne's participatory budget, and Swiss cities such as Aarau and Winterthur that have already used the Method of Equal Shares. I will partner with them as I do through Science for Democracy.

**Thrust C: A geometry of collective decision problems, from ballots to text.**

- **Computing invariant distances at scale.** Which domains admit polynomial-time relaxations of isomorphism-invariant distances that, like the positionwise distance, preserve the geometry? Can sketching and sublinear techniques make them scale to elections with millions of voters? This is the first PhD project of the thrust.
- **Benchmarks with provable coverage.** How many instances, and which ones, must a benchmark contain to cover the space of instances up to a given precision under a given distance? Which samplers achieve this? The answers will become defaults in `prefsampling` and `mapel`.
- **Maps of deliberation.** Extending the geometry from ballots to sets of textual opinions will make it possible to compare deliberations, human or AI, the way we now compare elections. This will give Thrust A its evaluation backbone.

**Students, funding, and service.** The three thrusts offer PhD projects of different flavors: theoretical (C), empirical and NLP-oriented (A), and applied and data-centric (B). I have already co-supervised more than ten students, led a grant as principal investigator (NCN Preludium), and worked within an ERC-funded project. I will apply to the Swiss National Science Foundation early on and for an ERC Starting Grant built around Thrust A. I serve on the editorial board of JAIR (since 2026) and on the program committees of NeurIPS, AAAI, AAMAS, and IJCAI.

## Why EPFL

My agenda sits between algorithms, AI, and the social sciences, and EPFL is one of the few places where all three are strong under one roof. I see concrete joint projects:

- with *Michael Kapralov* on sketching and sublinear algorithms for isomorphism-invariant distances between very large elections (Thrust C);
- with *Robert West* on how faithfully LLMs represent the opinions of real populations, and on polarization dynamics in AI societies, combining his work in computational social science with my preference measures (Thrust A);
- with *Antoine Bosselut* on measuring representation in LLM-generated summaries and consensus statements (Thrust A);
- with *Bryan Ford* on secure, auditable platforms for participatory processes and digital deliberation (Thrust B).

Beyond EPFL, the Lake Geneva region offers what this agenda needs most: direct democracy, municipalities that experiment with participation, and international organizations whose negotiations are rich records of real deliberation.